Chicago Housing Risk Analysis:

Economic and Housing Structural Determinants of Building Violations

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Research Question

- **How are housing violations distributed across neighborhoods in Chicago?**

  Spatial variation

- **To what extent are these patterns associated with economic pressure and housing structural features?**

  Economic Pressure

  - Rent burden
  - Median household income

  Housing Structural Features

  - Poverty rate
  - Ownership status
  - Building age

Data Resources

- https://data.census.gov/table (American Community Survey, ACS)

File name	Content
rent_burden_B25070.csv	Gross rent as a percentage of household income in the past 12 months
median_household_income_B19013.csv	Median household income in the past 12 months
poverty_status_B17001.csv	Poverty status (for calculating poverty rate)
tenure_owner_renter_B25003.csv	Renter and owner counting (renter-occupied percentage)
units_in_structure_B25024.csv	Building age classified by decades

- https://data.cityofchicago.org/Buildings/Building-Violations/22u3-xenr/data_preview (City of Chicago Data Portal)

building_violations_raw.csv	Building violation record

Data Progress – Clean Data

- Time window filtering: restricted records to 2020–2024
- GEOID Standardization: unified 11-digit census tract GEOIDs
- Column & Label Cleaning: unified column names and simplified labels
- Variable Selection (Estimate Only): retained ACS Estimate values and removed Margin of Error columns
- Missing Data Handling: checked for and cleaned missing/invalid (NaN) values before and after merging

Data Progress – Dependent Variable (Building Violations)

Violation Rate Construction

- Converted violation counts into a rate per 1,000 units:

  rate = 1,000 × (count / total units).

- After constructing the tract-level violation rate, we aggregated it monthly at the tract level to create a time series.

Data Progress – Independent Variable (Economic pressure & Housing Structural Features)

Standardization (by z-scores, using $z = \frac{x-\mu}{\sigma}$)

- Economic pressure Index:

  rent_burden_z-index – median_income_z-index

- Housing Structural Index:

  poverty_z-index + tenure_z-index + housing_age_z-index

  (note: Here, tenure is measured as the percentage of renter-occupied units in each census tract, rather than tenancy term. Higher renter-occupied share is expected to correlate with higher building violation rates since: Landlords may have weaker incentives for regular maintenance of rental properties compared to owner-occupied homes; Short-term rental tenancy reduces tenants' motivation to report minor violations before they escalate, leading to accumulated housing code violations.)

Derived Data

File name	Content
economic_pressure.csv	tract-level measures of median household income and rent burden used to capture economic housing pressure
housing_structural_features.csv	tract-level indicators of poverty status, tenure (owner vs. renter), and housing structure characteristics to measure structural vulnerability
building_violation_by_code_rate.csv	tract-by-month (and by violation code) measures of building violations, including raw counts and a standardized violation rate per 1,000 buildings. Each row corresponds to a census tract (GEOID), a month, and a violation code/description

Weakness and Difficulties

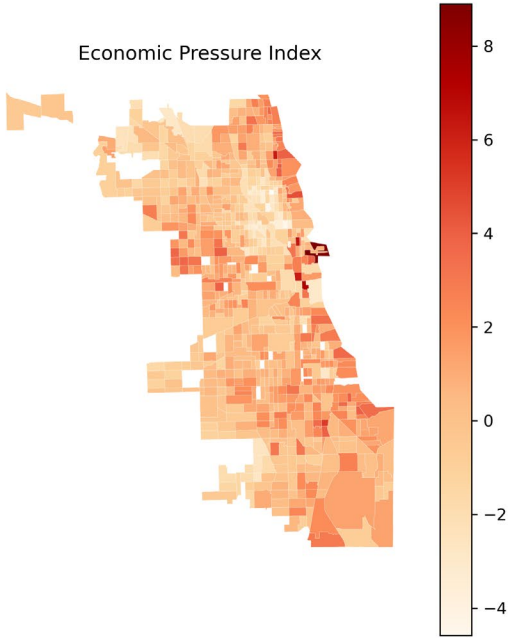
- Relative rather than absolute measure:

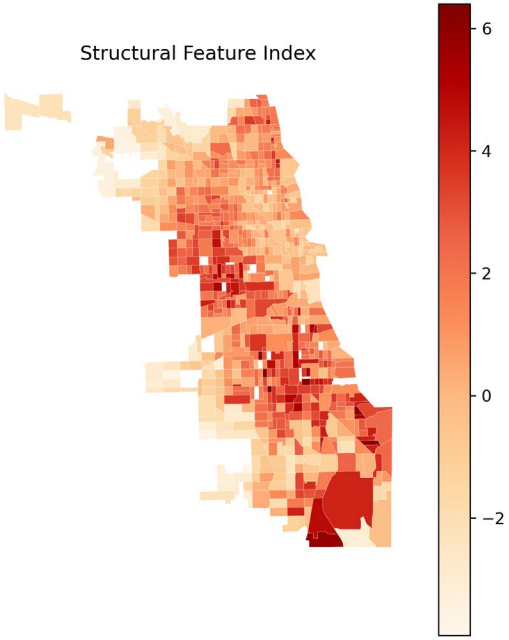
  Because we use z-score standardization, the index captures relative differences within Cook County rather than absolute levels of housing pressure.

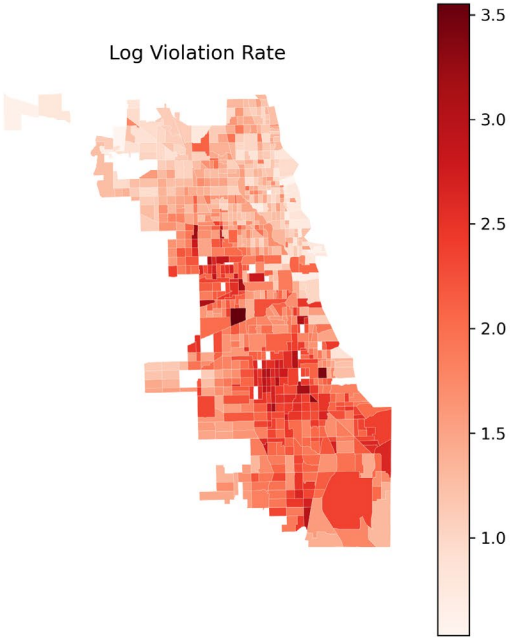
- Equal weighting assumption:

  By aggregating standardized variables, we implicitly assume equal contribution of each component, although their real-world impacts on housing pressure may differ.

Static Visualization – Spatial Distribution







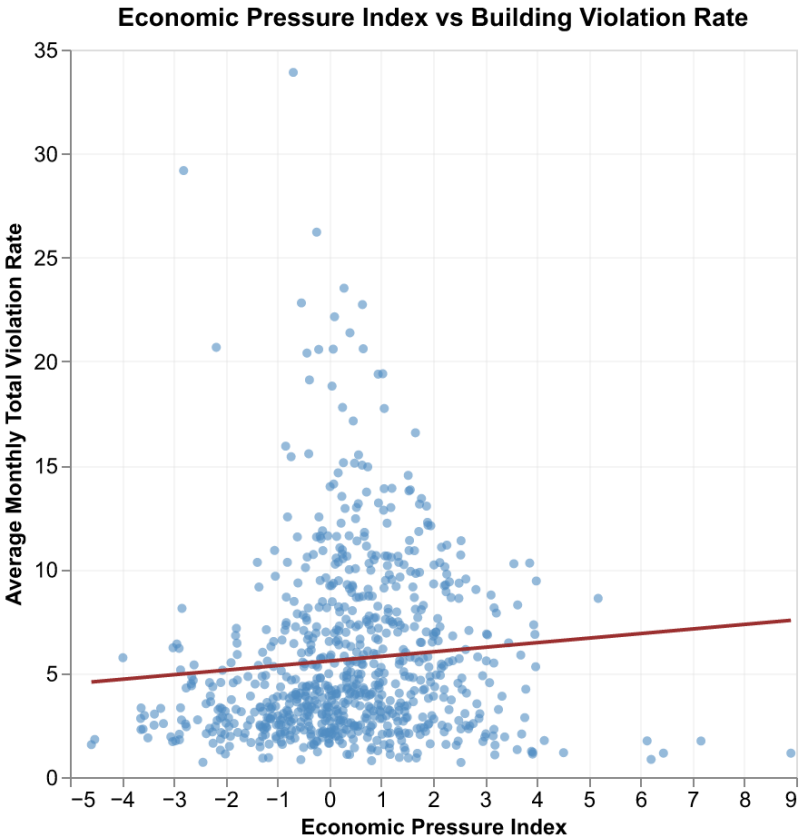
- Uneven but diffuse
- Higher values appear across multiple areas

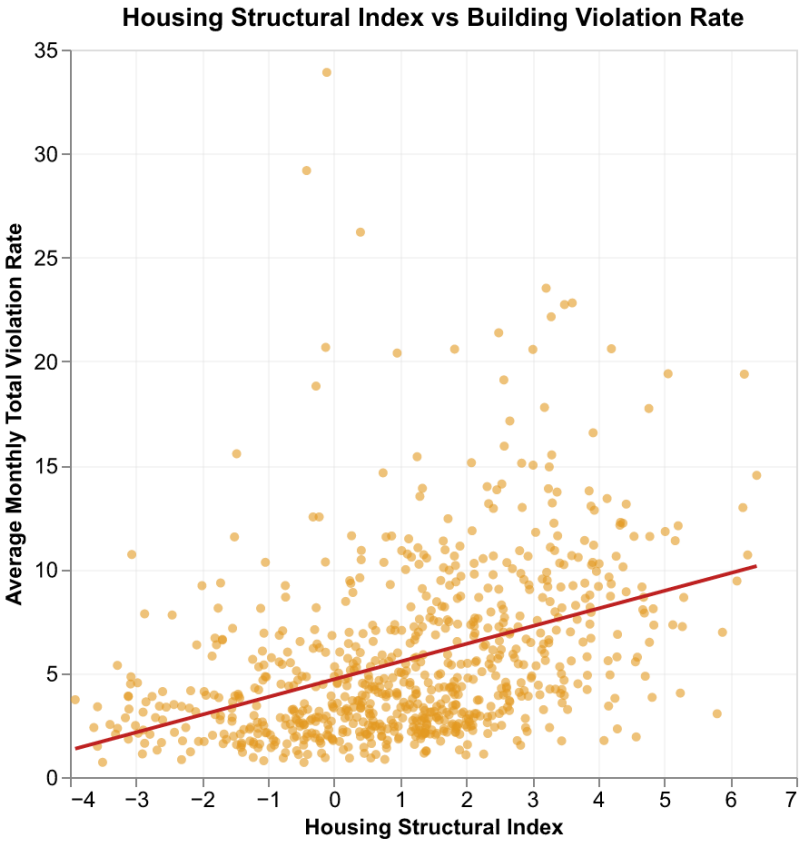
- Clearly spatial cluster
- Concentrated in the south/southwest

- Strong spatial cluster
- Concentrated in the south/southwest

- **Structural index shows a stronger spatial alignment with violation rates**
- **Economic pressure appears less closely matched**

Static Visualization – Correlation Analysis





While both factors show a positive association with violations, the relationship is stronger and more consistent for the Housing Structural Index.

Static Visualization – Correlation Analysis

Pearson Correlation:

- Economic Pressure Index vs Average Monthly Total Violation Rate:

  Correlation coefficient = 0.0840

  p-value = 0.019367

- Housing Structural Index vs Average Monthly Total Violation Rate:

  Correlation coefficient = 0.3911

  p-value = 0.000000

	economic_pressure_index	housing_structural_index	avg_monthly_total_violation_rate
economic_pressure_index	1.000	0.5360	0.0840
housing_structural_index	0.536	1.0000	0.3911
avg_monthly_total_violation_rate	0.084	0.3911	1.0000

The weak influence of economic pressure on the rate is likely to be indirectly generated through the mediating variable of housing structural features.

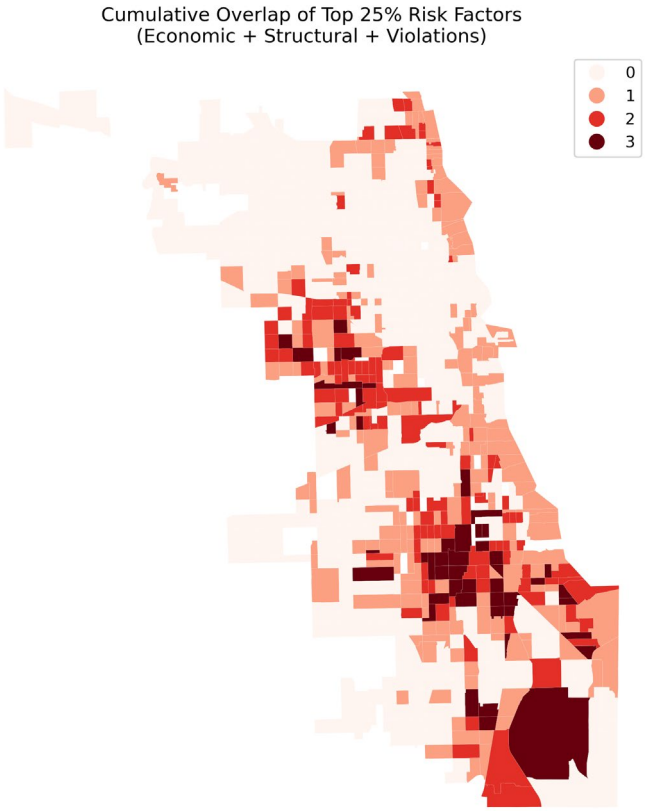
Dashboard

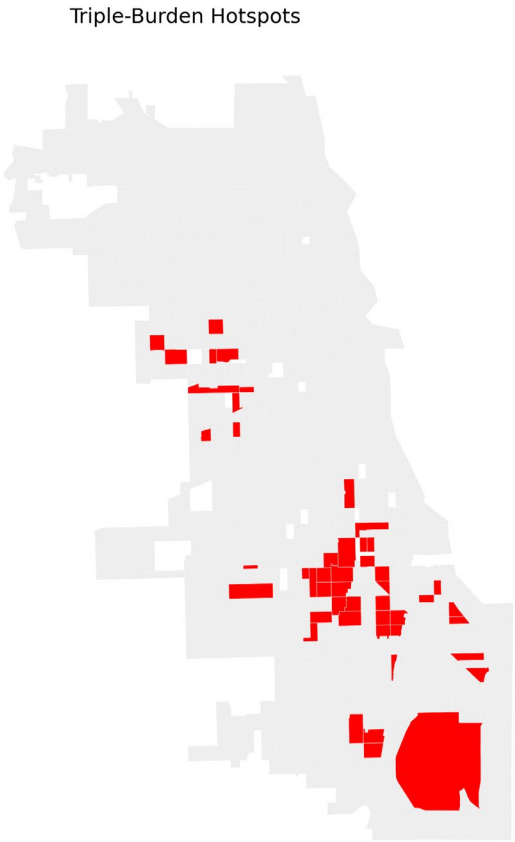
Streamlit Link: https://finalproject-susanweiyiyuqian-lq3bmd7kuccd7dhsba9gcz.streamlit.app/

We developed an interactive Streamlit dashboard to explore the distribution of housing violations across Chicago census tracts. The dashboard allows users to visualize violation rates, inspect tract-level information, and filter violations by category and time. It also includes a hotspot overlap tool that identifies tracts in the top percentile of economic pressure, structural vulnerability, and violation rates. These exploratory visualizations help identify spatial concentrations of risk and motivate the subsequent analysis of spatial patterns and overlapping housing risks.

Spatially, violation rates show persistent within-city imbalance: hotspots repeatedly cluster in Chicago's southwest corridor, while north-side and lakefront areas remain relatively low.

Static Visualization – Policy Extension





① Identify top 25% of three indicators
② Recoded into binary indicators
(1 = high risk;
0 = otherwise)
③ Accumulated risks and spatial overlap

- These hotspots face multiple overlapping housing risks and should be prioritized for policy intervention

Connection to Policy

Key Findings:

- Violations are spatially concentrated on the West and South sides.
- A gradual downward trend follows in subsequent years.
- Structural vulnerability shows a stronger association with violations ($r = 0.39$) than economic pressure ($r = 0.08$).
- High-risk tracts cluster geographically, forming triple-burden hotspots.

Policy Recommendations:

- The results suggest that housing policy should prioritize targeted inspections and maintenance programs in these high-risk neighborhoods. Because structural vulnerability shows a stronger association with violations than economic pressure, interventions focused on improving building conditions such as proactive inspections and rehabilitation support may be particularly effective. By using spatial risk mapping to identify and prioritize hotspots, housing agencies can allocate limited enforcement and maintenance resources more efficiently.

Thank you!

Susan, Weiyi and Yuqian

(Revised after presentation according to feedback)